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Comprehensive Investigation of Transient Joule-Thomson Cooling and CO₂ Hydrate Formation in Depleted Gas Reservoirs

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Abstract

Storing carbon in depleted gas reservoirs is crucial for mitigating global carbon dioxide (CO₂) emissions. Injecting high-pressure CO₂ into low-pressure reservoirs results in Joule-Thomson (JT) cooling caused by gas expansion. This temperature drop can lead to pore water freezing or the formation of CO₂ hydrates, potentially affecting injectivity and compromising well integrity. These processes remain poorly understood due to their complexity and non-linear behavior.

In this study, we present a coupled model that simulates non-isothermal fluid flow, hydrate formation kinetics, and the associated equations of rock properties. We verified this model using both steady-state and transient analytical solutions to analyze the dynamics of pore pressure propagation, thermal fronts, and fluid properties. Our results show that the developed model accurately captures the dynamics of these variables. Furthermore, we validated the model against experimental results, and it effectively predicts the spatial and temporal variations in pore pressure dynamics, both before and after the formation of CO₂ hydrates.

We applied the model to a field case (Viking gas reservoir), which has been considered for CO₂ storage. We presented field test cases with injection temperatures (T_{inj}) of 75°C, 50°C, and 25°C. The analysis of the gas reservoir provided critical insights into pressure and temperature evolution, especially under the different injection temperatures for each test case. Notably, when T_{inj} = 25°C, all test cases experienced the formation of CO₂ hydrates in the highly permeable layers, significantly affecting pressure gradients both before and after encountering the permeability barrier. Therefore, the developed non-isothermal flow model, which incorporates hydrate formation kinetics, can be utilized in complex heterogeneous and simple homogeneous models to forecast the dynamics of physical processes during the injection of CO₂ into depleted reservoirs.

Introduction

The Earth's climate has experienced substantial changes due to global warming, significantly affecting humans and the environment. The primary cause of global warming is the excessive release of CO₂ gas originated from fossil fuels (Rose et al., 2017). Governments around the world are taking proactive steps and actively seeking effective strategies to mitigate the effects of global warming. One potential approach is to lower atmospheric CO₂ levels through carbon capture and storage by injecting CO₂ into depleted petroleum reservoirs and saline aquifers (Hamieh et al., 2022; IPCC, 2005; Rowaihy et al., 2025). These underground formations act like a net-zero carbon sink, offering potential solutions for a sustainable future.

Depleted petroleum reservoirs in sedimentary basins have shown significant potential for CO₂ storage (Bui et al., 2018; Ye et al., 2023). These reservoirs, well-characterized by oil and gas operations, provide safe storage zones due to considerable pressure margins. Post-exploration and production operations, these reservoirs, characterized by low pressure, have proven seal capacity and containment. Depleted reservoirs in sedimentary basins typically occur within 700–4500 m depth from the surface. Deep saline aquifers fall within the same depth range as depleted reservoirs in sedimentary basins with high initial pressure and temperatures (Fig. 1). The depleted reservoirs benefit from existing basin information, the possibility of reusing infrastructure, and low pressure in the reservoir, which aids in keeping the CO₂ in the gas phase (Fig. 1). Moreover, the safety of these reservoirs for CO₂ storage provides reassurance against potential risks. However, the large-scale injection of CO₂ into depleted reservoirs is an area that requires further exploration and comparison with the benefits of storing in deep saline aquifers.

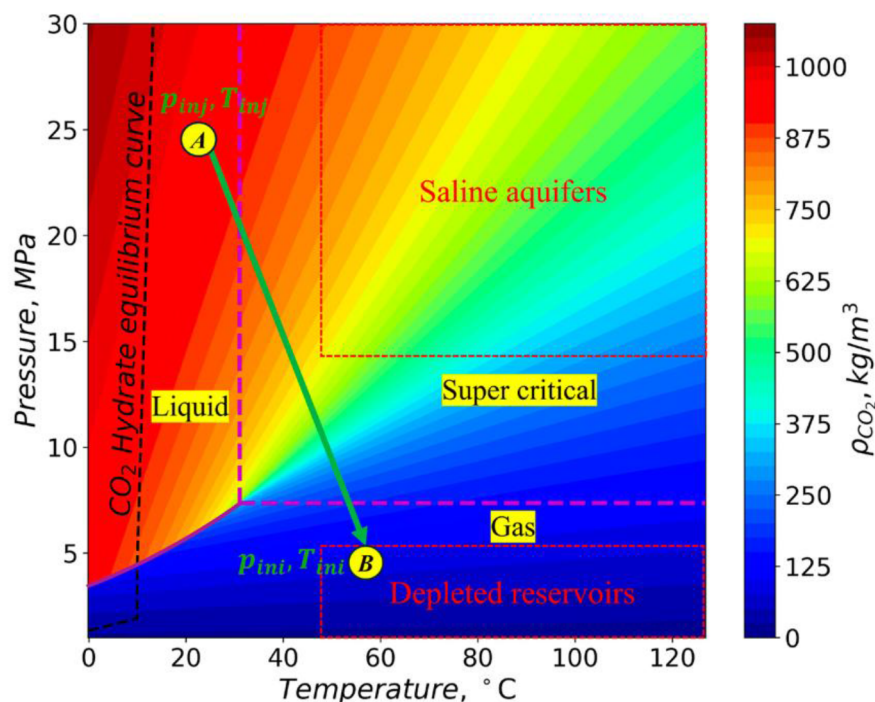


Figure 1—Pressure-temperature phase diagram of CO₂ showing the initial reservoir and injection conditions with phase separations. The density of CO₂ is collected from the CoolProp (<http://www.coolprop.org>), a Python library (Bell et al., 2014)

Pressure depletion enhances the pressure margin for higher CO₂ storage capacity; however, injecting high-pressure CO₂ into low-pressure formation lowers the reservoir temperature where fluid expansion occurs at constant enthalpy, which leads to various technical challenges. This effect is known as the Joule–Thomson (JT) cooling effect. It describes adiabatic cooling when a fluid expands and depressurizes across a pressure gradient. Although this cooling is localized, it can lead to thermal fracturing and further formation of CO₂ hydrate by forming a cage structure with the formation water. Furthermore, it may clog the pore

space and reduce the effective permeability. This can increase the pore pressure and the risk of injectivity loss and reservoir integrity. To mitigate this risk, CO_2 could be injected in its gaseous phase or get preheated to reduce the Joule-Thomson cooling effect. This methodology, however, is costly and carbon-intensive, as experienced in the North Sea Porthos CO_2 Project (Neele et al., 2019). To ensure safe and large-scale storage in depleted reservoirs, assessing the risks associated with JT cooling and developing innovative mitigation strategies to facilitate the injection of cold CO_2 into these reservoirs is essential.

The Joule-Thomson (JT) cooling effect significantly enhances the formation of CO_2 hydrate near the injection well. However, most studies have ignored hydrate formation kinetics (Akmal et al., 2024; Mathias et al., 2010; Tweed et al., 2024). Gas hydrates are crystalline compounds that consist of small gas molecules, such as methane (CH_4) or CO_2 , along with a specific number of water molecules (Sloan Jr & Koh, 2007; Zatsepina & Pooladi-Darvish, 2011). When high-pressure and low-temperature conditions are present in a reservoir, water molecules rearrange into crystal-like structures, creating cages that trap gas molecules. During the formation of CO_2 hydrate in the subsurface, the formation water acts as the host molecule, while CO_2 (or methane, if injected instead) serves as the guest molecule. CO_2 hydrate forms under conditions of low temperature and high pressure, as illustrated by the equilibrium curve for CO_2 hydrate in Fig. 1 (Tweed et al., 2024). The process begins when the reservoir pressure and temperature fall within the hydrate formation zone (HFZ), indicated in Fig. 2 (i.e., $p > p_{e,\text{CO}_2}$). As CO_2 injection into the reservoir continues, the JT cooling effect becomes more pronounced, causing the reservoir temperature to drop below the hydrate-forming temperature (HFT) and initiating the nucleation of the hydrate. Over time, these hydrates develop a crystalline structure, and the hydrate crystals may aggregate, leading to pore clogging, which affects permeability. The reduction in permeability of the rock matrix is proportional to the rate of hydrate formation. If hydrates form rapidly and block the pores, the permeability will decrease quickly, resulting in an increase in the injection pressure (from the injection point to the location of hydrate formation). This ultimately lowers the injectivity of CO_2 . Therefore, understanding and accurately modeling hydrate formation kinetics is crucial during the injection phase of CO_2 sequestration in depleted reservoirs.

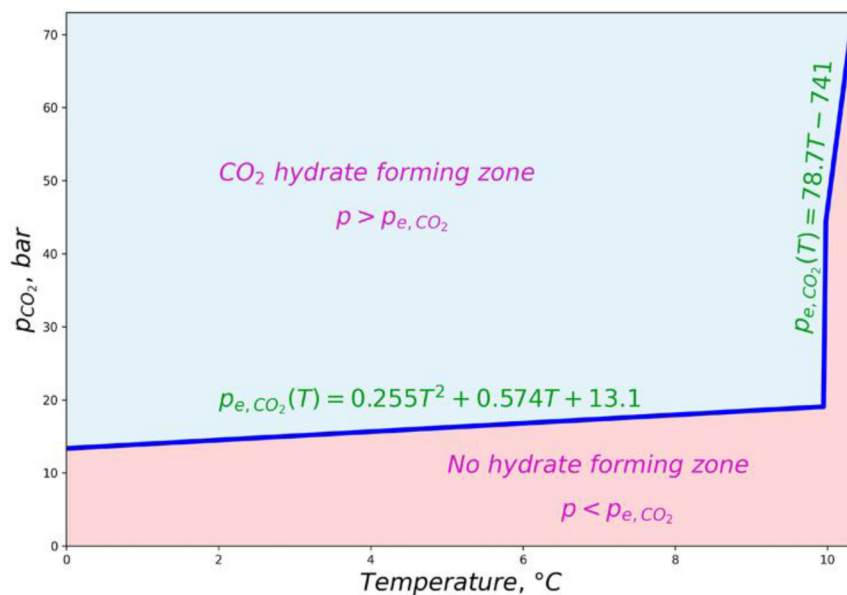


Figure 2— CO_2 hydrate-forming conditions (Akmal et al., 2024)

Several researchers have developed simulation models to explore the formation of CO_2 hydrates in porous media (Idress et al., 2018; Singh et al., 2020; Z. Sun et al., 2016). Most simulation models related to CO_2 hydrates focus on either gas production through depressurization or heating (X. Sun et al., 2005; Z. Sun et

al., 2016), or CO_2 sequestration in the form of hydrates by producing methane (Singh et al., 2020; Uddin et al., 2008). These models typically integrate thermo-hydrodynamics with hydrate formation kinetics but often neglect the JT cooling effect and gas expansion from liquid or supercritical states. The present study aims to couple both the JT cooling effect and hydrate formation kinetics with the flow model to analyze the impact of JT cooling on CO_2 hydrate formation, an aspect that has not been previously modeled in depleted reservoirs.

Joule–Thomson cooling effect in porous media is numerically modeled initially by Oldenburg (2007), who demonstrated that a pressure differential of about 5 MPa between an injection well and a reservoir boundary could lower temperatures by as much as 20°C. However, the author concluded that the Joule–Thomson cooling effect would unlikely initiate water freezing or CO_2 hydrate formation in the porous media. The study did not address the transient effects and the effective thermal conductivity. André et al. (2010) investigated the thermal impact of supercritical CO_2 injection with geochemical reactions in saline aquifers. They considered the Joule–Thomson cooling effect, water evaporation, and exothermic heat due to the dissolution of minerals. They found that the Joule–Thomson cooling effect significantly impacts the flow dynamics in saline aquifers and depleted gas reservoirs. Creusen (2018) studied the impact of CO_2 injection in a depleted gas reservoir. The author investigated the impact of CO_2 injection in the P18-4 gas field (North sea Parthos CO_2 project) which is partially depleted. The author found that the operating conditions used in the simulation would not favor the formation of CO_2 hydrate in that field. Mathias et al. (2010, 2014) developed an analytical solution using a steady-state approach and compared it with the TOUGH-2 numerical simulator. The authors ignored the influence of the thermal conductivity of both fluid and rock matrix. Akmal et al. (2024) developed a machine-learning model to predict the Joule–Thomson cooling effect in depleted gas reservoirs. They used the steady-state model to generate simulation data for the machine-learning model. They ignored the effects of thermal conductivity and considered only the heat transfer due to flow. The assumption of steady-state condition and fixed fluid properties are not consistent with the transient response during the injection of CO_2 into the depleted gas reservoir in the early injection period. Tweed et al. (2024) developed an analytical model for both steady-state and transient conditions to address the effects of Joule–Thomson cooling. They considered the dynamics of fluid properties in the transient analytical solutions. They treated the injection rate as the injection boundary condition, which is later converted into injection pressure with assumption of pseudo-steady state in the reservoir. These limitations from the aforementioned literature will provide unrealistic results of the simulation studies.

In this study, we develop a mathematical and numerical model to investigate the flow issues that arise during the injection of CO_2 into depleted gas reservoirs. This work is inspired by the contributions of (Maloney & Briceno, 2009) and Hoteit et al. (2019). High-density CO_2 (either in liquid or supercritical state) is injected into the depleted gas reservoir at high pressure. This leads to the rapid conversion of CO_2 from the liquid or supercritical state to gas. As a result, the density and viscosity of the injected CO_2 drop significantly, which increases the flow velocity. Additionally, the expansion of CO_2 causes the JT cooling effect, resulting in a localized temperature drop within the reservoir (Akmal et al., 2024; Mathias et al., 2014; Oldenburg, 2007). When the temperature falls below the hydrate formation threshold, and in the presence of water and CO_2 gas, CO_2 hydrate will form. This process can block the pores and reduce permeability, subsequently decreasing injectivity (Hoteit et al., 2019; Singh et al., 2020; Sloan Jr & Koh, 2007). Our model captures these complex physical phenomena through a coupled thermo-hydraulic framework that accounts for the Joule–Thomson cooling effect and localized hydrate formation kinetics.

The model incorporates hydrate formation kinetics, and the relevant equations become active when the reservoir temperature reaches or falls below the hydrate formation temperature (HFT). The localized nature of the Joule–Thomson cooling effect facilitates hydrate formation within the same area of the porous media. Therefore, saturation equations for hydrate and water are modeled separately, without including an advection term, and are triggered by the formation of CO_2 hydrate. This model integrates fluid properties

as functions of pressure (p) and temperature T, while rock properties are considered functions of hydrate saturation (S_{hg}), reflecting localized variations in porosity and permeability. We rigorously validate the model against analytical solutions and experimental results. Furthermore, we apply the mathematical model to the Viking depleted gas reservoir in the North Sea, testing three different scenarios: (i) injection temperature (T_{inj}) equal to the initial reservoir temperature (T_{ini}); (ii) conditions where $T_{f,hyd} < T_{inj} < T_{ini}$; and (iii) injection temperature near the hydrate formation temperature ($T_{f,hyd}$). We also investigate the sensitivity of injection pressure and boundary conditions during the CO_2 injection process, as well as the impact of reservoir domain size and associated boundary conditions.

Mathematical modeling

Governing equations

This section provides a detailed description of the mathematical model. The flow of CO_2 is governed by the conservation of mass and conservation of momentum, as presented in Eq. 1.

$$\frac{(\partial \phi_m \rho_g S_g)}{\partial t} + \nabla \cdot \left[\rho_g \left(\frac{\kappa_m}{\mu_g} \nabla p \right) \right] = -\hat{m}_{hg} \left(\frac{M_g}{M_{hg}} \right) \quad (1)$$

In Eq. 1, ϕ_m is the rock porosity, ρ_g is the density of CO_2 , S_g is the gas saturation, t is the time, κ_m is the rock permeability, μ_g is the viscosity of CO_2 , p is the pressure, $\hat{m}_{hg}$ mass rate of hydrate, M_g is the molecular weight of gas, M_{hg} molecular weight of CO_2 hydrate. The mass source term $-\hat{m}_{hg} \left(\frac{M_g}{M_{hg}} \right)$ represents the hydration formation during the fluid flow in porous media. This term takes effect when the hydrate formation conditions are met. The full expanded for of Eq. 1 gives Eq. 2.

$$\rho_g \beta_g S_g \frac{\partial p}{\partial t} - \frac{\rho_g \kappa_m}{\mu_g \phi_m} \nabla p + \frac{\rho_g \kappa_m \alpha_g}{\mu_g \phi_m} \nabla T \nabla p = \frac{\rho_g \kappa_m}{\mu_g \phi_m} \beta_g (\nabla p)^2 + S_g \rho_g \alpha_g \frac{\partial p}{\partial t} - \hat{m}_{hg} \left(\frac{M_g}{M_{hg}} \right) \quad (2)$$

In Eq. 2, β_g is expansion coefficient of CO_2 , α_g is thermal expansion coefficient of CO_2 , T is the temperature. Both β_g and α_g are functions of pressure and temperature. The mass balance equation for immobile water at the hydrate formation location and mass balance equation for hydrate are presented in Eq. 3 and Eq.4. Both Eq. 3 and Eq. 4 will be activated automatically in the numerical simulations after satisfying the conditions as presented in Fig. 3.

$$\frac{(\partial \phi_m \rho_w S_w)}{\partial t} = -\hat{m}_{hg} \left(\frac{N_h M_w}{M_{hg}} \right) \quad (3)$$

$$\frac{(\partial \phi_m \rho_w S_w)}{\partial t} = \hat{m}_{hg} \quad (4)$$

The mathematical equations which govern the heat transfer in porous media while flowing the two phases were presented in the Eq. 5 to Eq. 10. This equation is coupled with two source terms: the first source term $\rho \nu \mu_{JT} C_p \nabla P$ represents the JT cooling effect due to pressure change and fluid expansion, while the second source term Q_{hyd} represents the exothermic nature of hydrate formation (Gudala et al., 2022; Singh et al., 2020; Tweed et al., 2024).

$$(\rho C_p)_{eff} \frac{\partial T}{\partial t} + \nabla \cdot (-\lambda_{eff} \nabla T) + \rho_g \nu C_{pg} \nabla T = \rho_g \nu \mu_{JT} C_p \nabla P + Q_{hyd} \quad (5)$$

$$(\rho C_p)_{eff} = \phi_m (S_g \rho_g C_{pg} + S_w \rho_w C_{pw} + S_{hg} \rho_h C_{ph}) + (1 - \phi_m) \rho_m C_{pm} \quad (6)$$

$$\mu_{JT} = \frac{\alpha_g \cdot T - 1}{\rho_g C_{pg}} \quad (7)$$

$$\alpha_g = \frac{1}{\rho_g} \left(\frac{\partial \rho_g}{\partial T} \right)_p \quad (8)$$

$$Q_{hyd} = \hat{m}_{hg} \Delta H_{hg} \quad (9)$$

$$\Delta H_{hg} = 2528T_{const}^8 + 75.36T_{const}^7 - 9727T_{const}^6 + 1125T_{const}^5 + 4000T_{const}^4 - 4154T_{const}^3 + 14430T_{const}^2 - 6668T_{const} + 389900 \quad (10a)$$

$$T_{const} = \frac{T - 278.15}{2.739} \quad (11b)$$

In Eq. (5) to Eq. (10), S_w is the water saturation, S_{hg} is the hydrate saturation, $(\rho C_p)_{eff}$ is the effective heat capacity, λ_{eff} is the effective thermal conductivity, μ_{JT} is the JT cooling coefficient, $C_{p,g}$ is the heat capacity of gas, $C_{p,w}$ is the heat capacity of water, $C_{p,h}$ is the heat capacity of CO_2 hydrate, v is the darcy velocity, ρ_m is the density of rock matrix, $C_{p,m}$ is the heat capacity of rock matrix, Q_{hyd} is the heat of hydrate formation.

Hydrate formation requires an adequate number of water molecules to encapsulate each gas molecule. Typically, under hydrate-forming conditions, one mole of hydrate-forming gas requires N_h moles of water to form a stable hydrate structure. The equilibrium reaction of formation and dissociation of CO_2 hydrate is presented in Eq. 11 (Singh et al., 2020).



In Eq. (11), N_h is the hydrate number. The hydrate formation rate is directly proportional to the pressure change between the hydrate equilibrium pressure and the gas phase pressure (Kim et al., 1987; Singh et al., 2020). Eq.12 represents the change of hydrate mass and associated equations are presented from Eq. 13 to Eq.15.

$$\hat{m}_{hg} = -K_{f,CO_2} M_{hg} A_{s,hg} (p_{e,CO_2}(T) - p) \quad (13)$$

$$K_{f,CO_2} = K_d^0 \exp\left(\frac{-\Delta E_a}{RT}\right) \quad (14)$$

$$M_{hg} = M_{CO_2} + N_h M_{H_2O} \quad (15)$$

$$A_{s,hg} = \sqrt{\frac{[\phi_m(1 - S_{hg})]^3}{2\kappa_m}} \quad (16)$$

In Eq (12) to Eq. (15), K_{f,CO_2} equilibrium constant of CO_2 hydrate, $A_{s,hg}$ is the hydrate forming area, $p_{e,CO_2}(T)$ is the equilibrium pressure, K_d^0 is the equilibrium constant at standard condition, ΔE_a is the change in energy, R is the universal gas constant. The formation of gas hydrate in the porous media may block the pore space due to the immobility of the hydrate at the hydration forming location. With the blocking of pore space, the permeability of the porous media will reduce and eventually may become impermeable. This reduction in porosity and permeability is included in Eq.16 and Eq.17, respectively (Qiu et al., 2023; Singh et al., 2020).

$$\phi_m = (1 - S_{hg}) \phi_{ini} \quad (17)$$

$$\kappa_m = \kappa_{ini} (1 - S_{hg})^{N_\kappa} \left(\frac{\phi_m}{\phi_{ini}} \right)^3 \quad (18)$$

In Eq.16, N_κ is the permeability reduction factor, ϕ_{ini} is the initial rock porosity, κ_{ini} is the initial rock permeability.

Analytical model

The analytical model is developed using the radial coordinate system, classical pressure diffusion equation is used to determine the injection pressure at constant temperature (Ahmed & McKinney, 2011; Dake, 1983).

$$\frac{\partial p}{\partial t} = \frac{\kappa_m}{\phi_m \mu_g \beta_f} \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial p}{\partial r} \right) \quad (19)$$

Eq. 18 is further modified using the constant terminal rate solution due to the constant injection rate at the injection well (Ahmed & McKinney, 2011; Dake, 1983). The Solution uses the exponential integral (i.e., $E_i(-x)$) to determine the pressure at the injection well with constant injection rate and is presented in Eq. 19 (Ahmed & McKinney, 2011; Tweed et al., 2024).

$$p_{wf}(t) = p_{ini} - \frac{\mu_g m_{inj,CO_2}}{4\pi H \kappa_m \rho_g} e^{\eta_w^2 E_i(-\eta^2)} \quad (20)$$

$$\eta(r, t) = r \left(\frac{4\kappa_m t}{\phi_m \mu_g \beta_f} \right)^{-1/2} \quad (21)$$

$$\eta_w(t) = r_w \left(\frac{4\kappa_m t}{\phi_m \mu_g \beta_f} \right)^{-1/2} \quad (22)$$

In Eq (19) to Eq. (21), m_{inj,CO_2} is the mass injection rate, r_w is the wellbore radius, p_{ini} is the initial reservoir pressure, H is the reservoir thickness, r is the radius of the reservoir. At typical operational conditions, the pressure drop near the injection well is reaches a pseudo-steady state (Ahmed & McKinney, 2011; Bear, 2012). The pressure gradient (∇P) in the pseudo-steady state regime with a constant mass injection rate is expressed in Eq. 22 (Mathias et al., 2010; Tweed et al., 2024)

$$\nabla P = - \frac{\mu_g m_{inj,CO_2}}{2\pi H \kappa_m \rho_g r} \quad (23)$$

Initially porous media is saturated with water (e.g., $S_{wr}=0.2$) and CH_4 (e.g., $S_{g,CH_4}=0.8$). As CO_2 is injected, it displaces the methane in the porous media at high injection pressure (e.g., 3500 psi). The injected fluid temperature (e.g., $T_{inj}=25^\circ\text{C}$.) is lower than the initial reservoir temperature. Eq. 23 and Eq. 24 represent the initial and boundary conditions,

$$p(x, t) = \begin{cases} p_{ini} & t = 0 \text{ or } x = L \\ p_{inj} & t > 0 \text{ and } x = 0 \end{cases} \quad (24)$$

$$T(x, t) = \begin{cases} T_{ini} & t = 0 \text{ or } x = L \\ T_{inj} & t > 0 \text{ and } x = 0 \end{cases} \quad (25)$$

Numerical model verification with the analytical model

The governing equations are simplified to match the assumptions considered in the analytical model. We consider the injection of CO_2 into incompressible system, which is initially saturated with CH_4 and immobile residual water. During the process, CO_2 is injected from the injection well, displacing CH_4 . For this simplified case, mixing between the gases is ignored. Furthermore, we assume there is no mineral precipitation or dissolution of CO_2 and CH_4 in the resident water. Therefore, this test mimics a single-phase fluid flow within the porous media.

The model accounts for the formation of CO_2 hydrates in the porous medium, as depicted in Fig. 1 and Fig. 2. The formation of these hydrates occurs when temperature and pressure fall within the appropriate conditions, as presented in Fig. 2. In this scenario, CO_2 hydrate formation is localized and immobile. The formation process takes into account the available water, which will decrease as the hydrate forms,

converting from liquid to solid state. Variations in porosity and permeability at these locations are modeled, as the pore space gets blocked by CO_2 hydrates. In this case, buoyancy and capillary forces are ignored.

Fig. 4 illustrates the schematic demonstrating the Joule-Thomson (JT) effect alongside the localized formation of hydrates influenced by boundary heating. The pressure profile (p) drops along the pathway from injection point A to boundary F along the path $A-B-F$, while the temperature (T) follows a non-monotonic profile, represented by $A-B-C-D-E$. The key mechanisms include advection (AB), JT cooling (BC), JT heating (CD), and the effects of boundary to heat and pressure (E and F). Within the BCD region, CO_2 expansion and phase-change takes place. The JT cooling effect leads to temperature drop in this area, impacting pressure and temperature profiles. This process can induce hydrate formation, which blocks pore space and decreases permeability. Consequently, this impacts the saturations of immobile water (S_w), gas (S_g), and CO_2 hydrate (S_h). The properties of CO_2 , including density, viscosity, heat capacity, thermal conductivity, and Joule-Thomson coefficients, are all functions of pressure and temperature are collected from the CoolProp library (Bell et al., 2014).

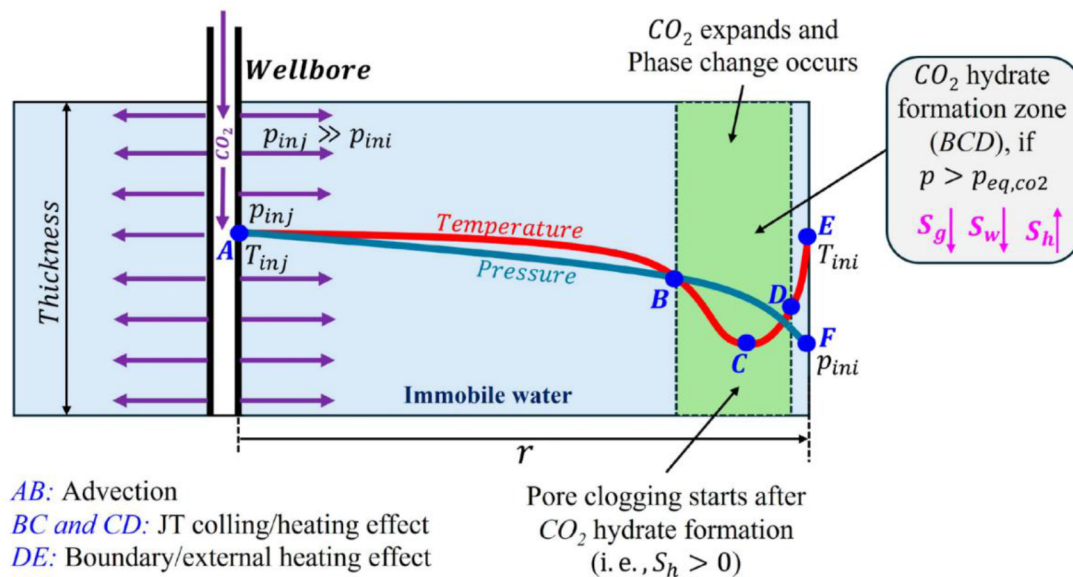


Figure 4—Schematic of Joule-Thomson cooling effect, CO_2 hydrate formation, and boundary heating effect during the injection of high-pressure CO_2 into a low-pressure reservoir.

Results and Discussions

Verification with steady state and transient state analysis

In this work, we verified our mathematical model with the analytical models both in the study and transient models. Tweed et al. (2024) used the non-dimensional approach to solve the governing equations with the boundary value approach for the transient state model. We used the COMSOL Multiphysics simulator to solve the governing equations (i.e., from Eq. 1 to Eq. 17). The schematic of the one-dimensional reservoir model used for the analytical models is presented in Fig. 5a. Initial and boundary conditions are also presented in Fig. 6a. The temporal variation of the injection pressure is calculated using the Eq. 19, as depicted in Fig. 7b.

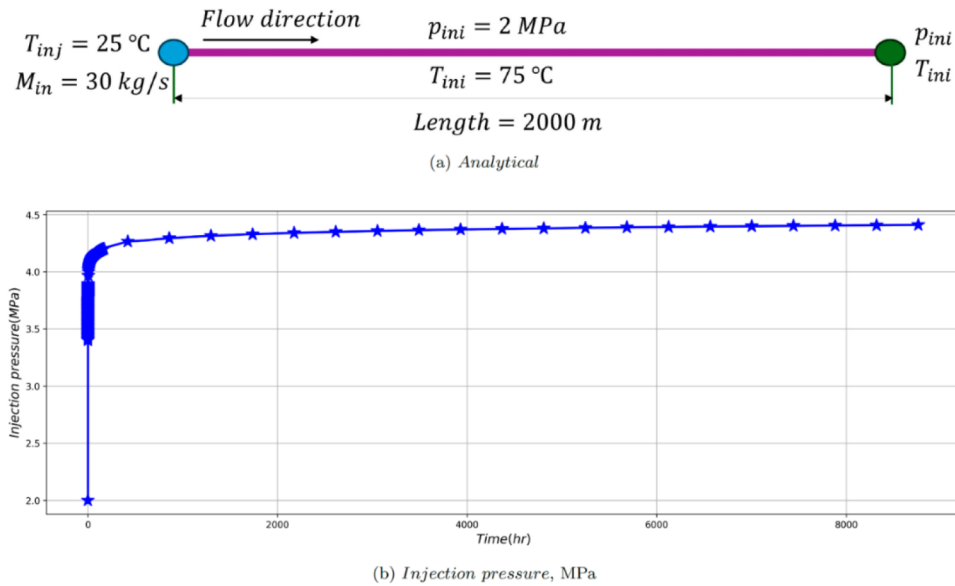


Figure 8—Schematic representations of the analytical model with initial and boundary conditions(a),and injection pressure(b)

A Python-based code has been developed to perform analytical solutions for both steady and transient state scenarios. The results of these analytical solutions, along with the developed model, are presented in Fig. 9. Over time, the pressure at the injection well increases, which is linked to the pressure observed at the injection site (refer to Fig. 10b). The pressure for steady-state and transient state exhibit different behaviors due to the constant density of CO_2 under steady-state conditions (i.e., ρ_{CO_2} at the injection well). At steady-state, other fluid properties are also constants at a given moment and depend on T_{inj} and p_{inj} . Conversely, in the transient scenario, fluid properties vary according to the spatial and temporal fluctuations of pressure (p) and temperature (T) within the reservoir. This variation is evident in the spatial changes of CO_2 density over time in the reservoir, as illustrated in Fig. 11.

We also examined variations in the JT, thermal, and fluid expansion coefficients, which depend on pressure and temperature. The simulation results for pressure, temperature, and CO_2 density generated by our model are illustrated in Fig. 12. Notably, the pressure profile aligns closely with the transient solutions of the analytical model, as seen in Fig. 13a. In contrast, the temperature profiles from our model reveal discrepancies between the steady-state and transient solutions of the analytical model, as shown in Fig. 14b. This variation stems from the dynamic nature of the coefficients μ_{JT} , α_g , and β_g . As indicated in Fig. 15b, the JT cooling effect contributes to the reduction in temperature, with cooling effects extending over time and causing the location of the lowest temperature to shift accordingly. We also investigated the spatial and temporal variations of ρ_{CO_2} within the porous media, discovering that ρ_{CO_2} is influenced by reservoir pressure and temperature. Consequently, the developed model is capable of analyzing the dynamic variations of fluid properties and the transient responses (i.e., pressure and temperature) that occur in the porous media.

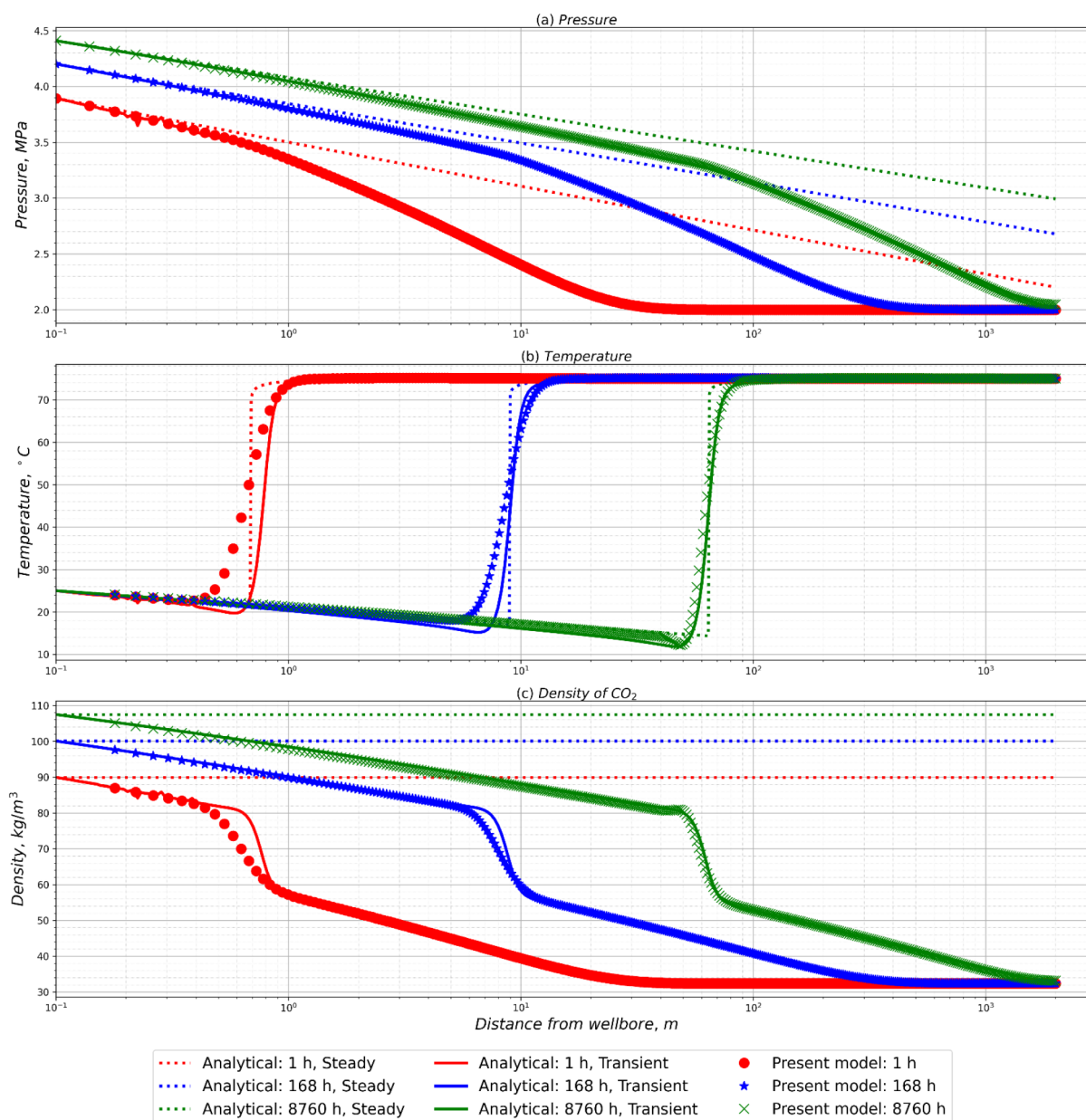


Figure 16—Pressure, temperature, density variations in porous media with different approaches of solving JT cooling effect in steady, and transient and developed mathematical model

Field study: Viking gas reservoir in the North Sea

The cooling effect associated with Joule-Thomson (JT) processes and hydrate formation has not been extensively investigated in real-field applications. To address this gap, we analyze the thermo-hydraulic processes in porous media alongside the kinetics of hydrate formation within a natural gas reservoir located at the Viking field in the North Sea, which has been considered for CO_2 storage (Hoteit et al., 2019; Tabrizinejadas et al., 2023).

Fig. 17 provides a three-dimensional view of the Viking field, a depleted gas reservoir exhibiting an immobile water saturation of 0.2. The injection location within the study area is also depicted in Fig. 17. The reservoir is comprised of 18 layers, each with heterogeneous properties. The JT cooling effect is most pronounced near the wellbore; thus, we employed a radial-coordinate system in this analysis. The layer-specific porosity and permeability characteristics of the study area are illustrated in Fig. 18a and Fig. 18b,

respectively. Porosity ranges from 0.106 to 0.2, while permeability varies from 30 md to 1250 md across the reservoir (see Fig. 18).

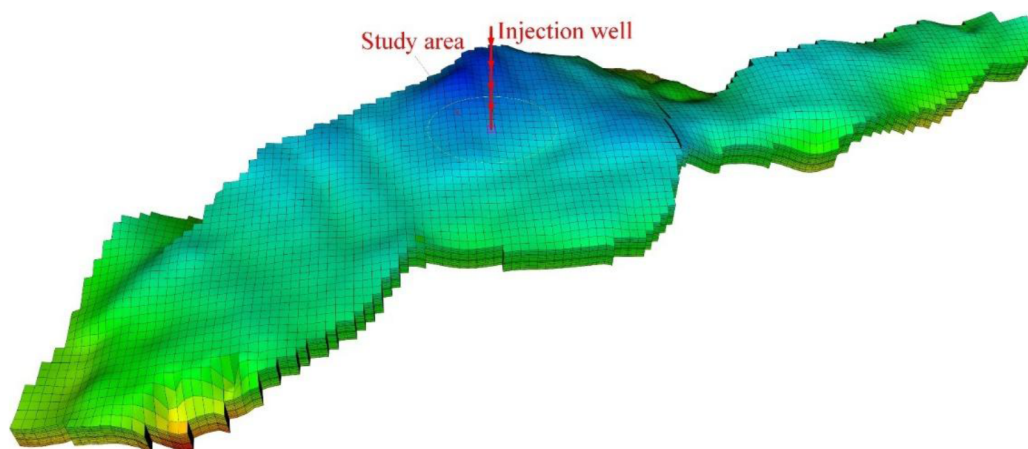


Figure 17—Geological model (grids) of Viking Field, showing injection well location and study area

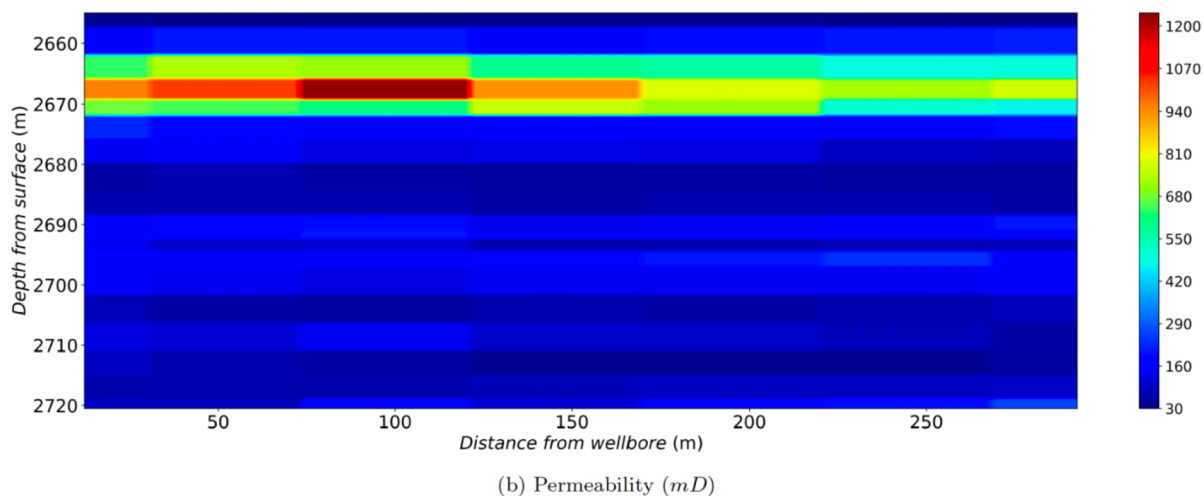
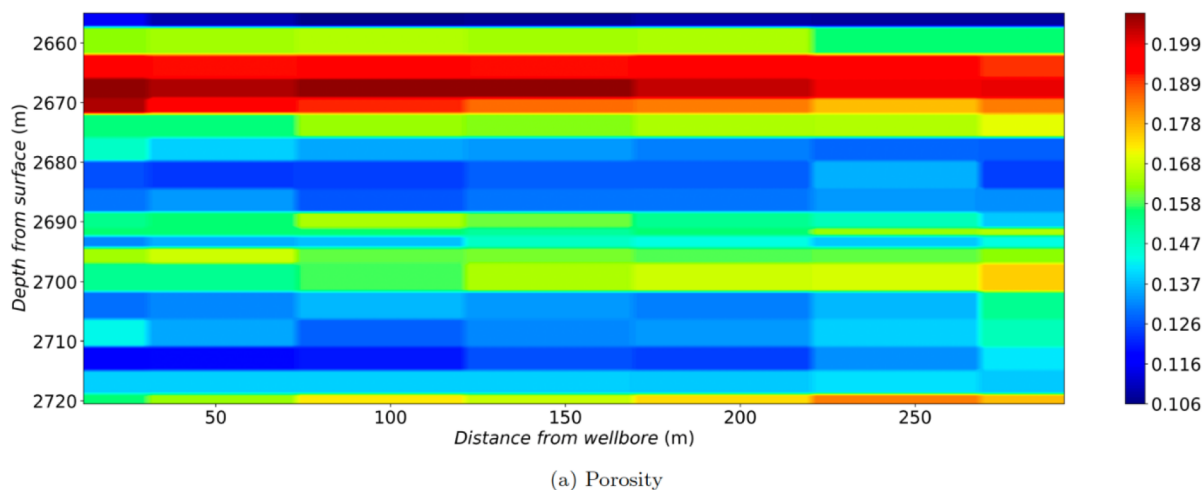


Figure 18—Distribution of porosity and permeability in a cross-section near the injection well.

We established two test cases to implement the JT-cooling effect, (1) model considering only high permeable layers; (2) model considering all layers of the reservoir and injecting CO_2 in high permeable

layers. For each test case, we analyzed three injection scenarios: (1) the injection temperature of CO_2 matches the initial reservoir temperature; (2) the injection temperature of CO_2 exceeds the hydrate formation temperature; and (3) the injection temperature of CO_2 is close to the hydrate formation temperature. At the depletion state, the pore pressure of the reservoir is 200 psi (1.38 MPa). The initial reservoir temperature is 75°C, with the injection temperatures for the three scenarios set at 75°C, 50°C, and 25°C, respectively. The injection pressure remains consistent across all three scenarios at 3500 psi (24.13 MPa). Table 1 presents additional properties utilized in the simulations.

Table 1—Properties of rock, fluid and CO_2 hydrate properties used in the present work

Property	Value
Rock Density (kg/m^3)	2600
Rock Thermal conductivity ($W/m \cdot K$)	2.0
Rock Heat capacity at constant pressure ($J/kg \cdot K$)	1000
Anisotropic ratio	0.1
K_o^d ($mol/(Pa \cdot s \cdot m^2)$)-Hydrate	0.35×10^5
N_h	5.75
ΔE_a (J/mol)	81084
$R_{(gas)}$ ($J/mol \cdot K$)	8.13
Hydrate density	1100
Hydrate heat capacity at constant pressure	2200

Fig. 19 illustrates the variation of pore pressure under three distinct injection temperature scenarios through the injection of CO_2 into the high permeability layers. The pore pressure ranges from 24.13 MPa at the injection well to 1.38 MPa at the boundary, demonstrating significant variation. Notably, the most pronounced changes in pore pressure occur near the wellbore during the early stages of injection, with pressure effects propagating outward over time (see Fig. 19).

Initially, elevated pressure is concentrated in the vicinity of the wellbore and exhibits a decrement with increasing distance and depth, as depicted in Fig. 19a(i). Subsequently, this high-pressure region extends further into the reservoir, thereby impacting a broader area (Fig. 19a(ii)). Over time, the pressure continues to propagate, leading to the development of more uniform pressure gradients across the reservoir, as shown in Fig. 19a(iii). Eventually, pressure stabilizes with observable evidence of dissipation at the reservoir boundaries (Fig. 19a(iv)). The pressure front advances (Fig. 19a(iv)). Ultimately, the system approaches a quasi-steady state, characterized by diminishing pressure gradients at the furthest distances (Fig. 19a(iv)). This propagation is primarily driven by the pressure differential between the injection pressure and the far field pressure boundary conditions in the sector model.

The pressure propagation observed in the $T_{inj} = 50^\circ C$ scenario aligns with previous findings from the $T_{inj} = 50^\circ C$ scenario (Fig. 19b). However, it diverges notably in the $T_{inj} = 25^\circ C$ scenario (Fig. 19c). In the initial time step (i.e., up to 10 days), pressure propagation mirrors that of the earlier scenarios at $T_{inj} = 75^\circ C$ and $T_{inj} = 25^\circ C$. Subsequently, pore pressure propagation leads to the establishment of two distinct pressure zones at approximately 110 m from the injection well (Fig. 19c(iii)). The formation of these two pressure zones is attributed to the filling of pore spaces with solid crystalline structures of CO_2 hydrates, which restricts the flow of injected CO_2 , functioning effectively as a barrier (Fig. 19c(iii) to Fig. 19c(iv)).

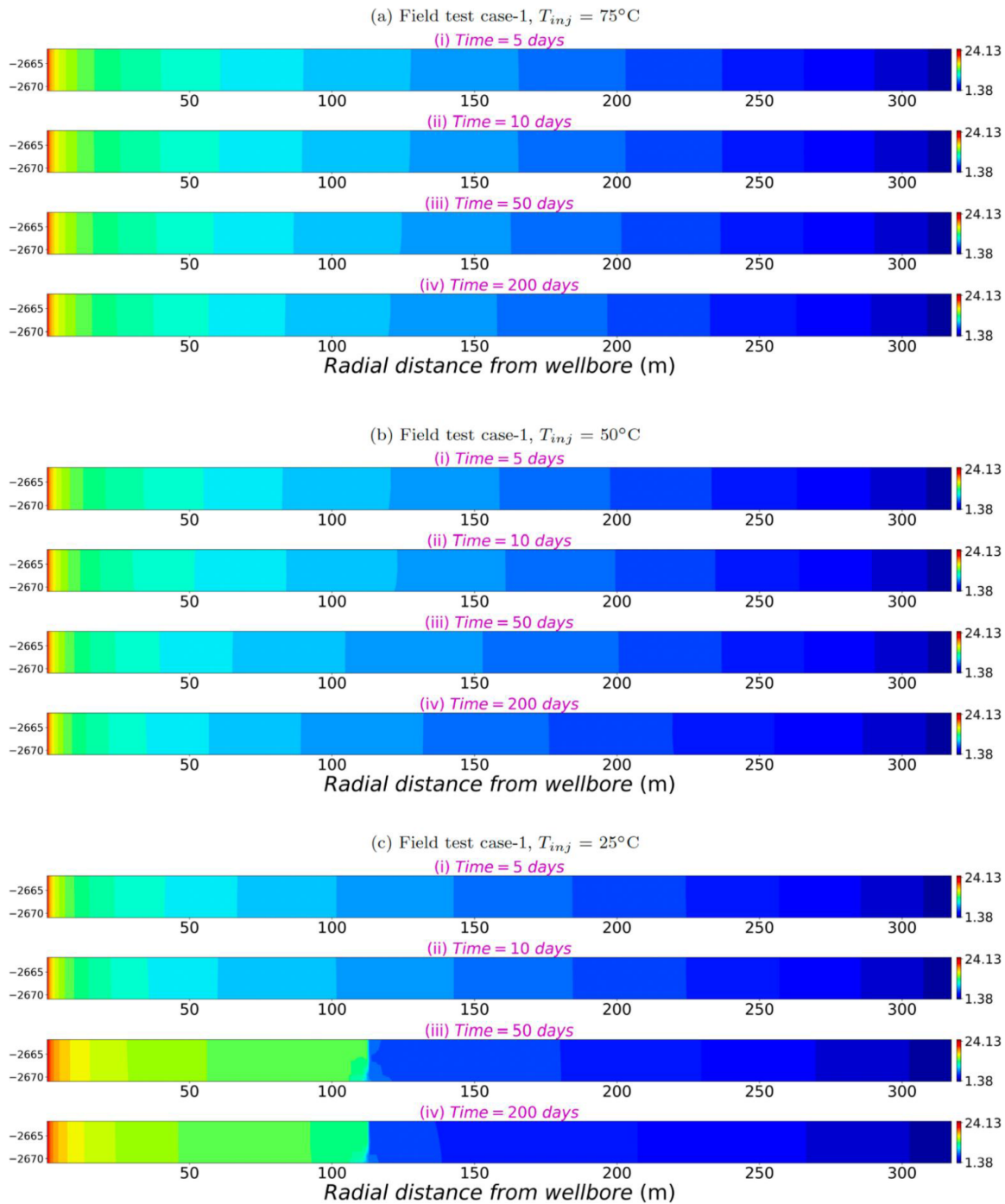


Figure 19—Variation of pore pressure (MPa) for field test case-1 with different injection temperatures.

As a result, the pressure gradient from the injection well to the barrier, as well as from the barrier to the outer boundary reduce, resulting in two different near-equilibrium conditions within the reservoir. Furthermore, only minimal vertical variations in pressure are observed between layers, attributable to the differences in layer-wise permeability within the reservoir. The developed model accurately predicts these variations, thereby contributing valuable insights into the dynamics of pore pressure evolution in this context.

Fig. 20a depicts the variation of temperature in the reservoir at three different injection temperatures. As the pressure drop increases across the domain, the temperature of the injected CO_2 reduces; further, it cools the rock matrix. In earlier steps, the temperature showed significant cooling near the wellbore while

injecting CO_2 at $75^\circ C$. This rapid cooling correlates with high-pressure gradients near the injection well (Fig. 20a). The significant drop in temperature during the early times suggests adiabatic cooling effects from CO_2 expansion as it transitions to lower pressures. This phenomenon is characteristic of JT cooling during CO_2 expansion. The low-temperature regime gradually propagates towards the outer boundary while advancing in time (Fig. 20a). The convective heat transfer is more prominent in this process than the conductive transfer. This is due to the dominance of fluid expansion and phase change due to heat exchange between rock-to-injected CO_2 and injected CO_2 -to-rock. The heating at the surrounding boundary maintains high temperature at the outer boundary.

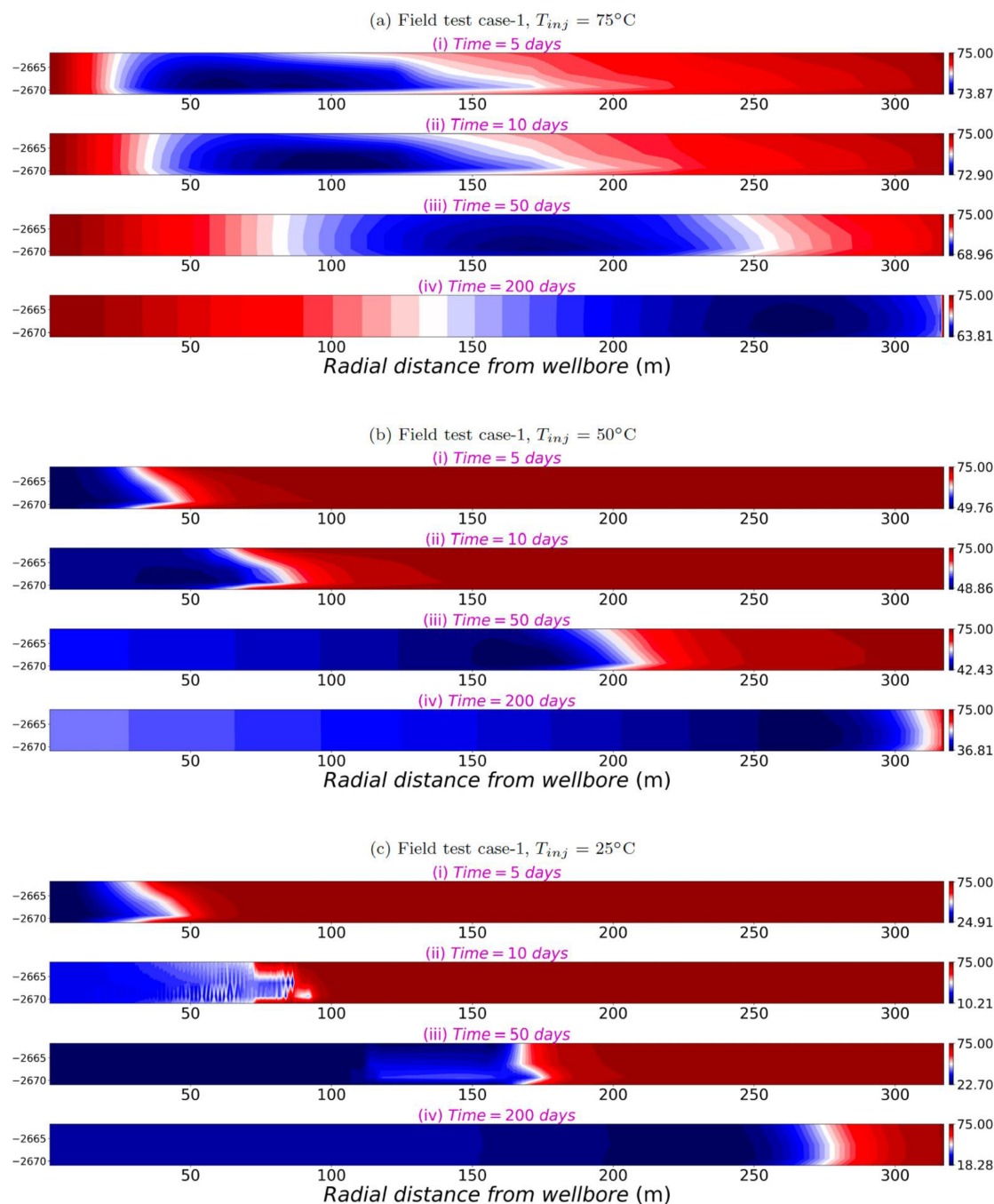


Figure 20—Temperature behavior for field test case-1 with different injection temperatures.

Fig. 20b depicts the temperature variation in the porous media while injecting CO_2 at 50°C. The low-temperature regime propagates towards the outer boundary from the injection wellbore. The fluid expansion is not very significant in the early times in this scenario. This is due to the preliminary heat exchange between the rock-to-injected CO_2 and injected CO_2 -to-rock. With time advances, conduction and JT cooling effects combined to cool down the rock matrix and injected fluid, leading to more significant temperature drops compared to scenarios-1 Fig. 20b(iii) to Fig. 20b(iv). This observation aligns with the previously analyzed pressure distribution of the same scenario (Fig. 20b). The lower injection temperature contributes to an active heat transfer between the reservoir rock and injected CO_2 and JT-cooling effect at higher pressure gradients.

Fig. 20c shows the temperature variation in the porous media while injecting CO_2 at 25°C. As the above scenario illustrates, conduction and JT-cooling effects work synergistically to bring the fluid temperature to hydrate formation levels. This temperature reduction (Fig. 20c(i) to Fig. 20c(ii)) creates a conducive environment for hydrate nucleation and growth. This temperature reduction creates a conducive environment for hydrate nucleation and growth. This hydrate uses the pore water and CO_2 in the pore to form in the same pore, close to the pore, and restrict the entry of incoming fluid by reducing the permeability. This will act as a barrier and create two pressure zones in the porous media. This is observed from the pressure analysis in Fig. 20c. Thus, the developed mathematical model effectively captures all the physical processes during the early stage of CO_2 injection into the depleted gas reservoirs. Those include the JT-cooling effect, the reservoir's convective and conductive heat transfer, and hydrate formation kinetics when the reservoir conditions fall within the CO_2 hydrate phase envelope.

Fig. 21 illustrates the variations in pressure and temperature after 150 days of CO_2 injection into high-permeability layers at different injection temperatures (T_{inj}). These plots are crucial for understanding how injection temperature affects the pressure and temperature distribution across the radial reservoir's spatial heterogeneities.

The pore pressure is observed to distribute radially from the wellbore to the outer boundary. In the high-permeable layers, pressure fluctuations in the vertical direction are minimal, whereas significant fluctuations occur between the high-permeable and low-permeable layers. In this scenario, the pressure gradients in high-permeable layers (specifically layers 3 to 5, which have a combined thickness of approximately 9.11 m or 30 ft) are higher than those in the low-permeable layers. This difference arises from the rapid transition of high-pressure CO_2 into the low-pressure reservoir, leading to a phase transition of the injected CO_2 , as illustrated by the transition from *A* to *B* in Fig. 1. This pressure behavior is consistently observed across all three scenarios of this test case, as shown in Fig. 21a, Fig. 21c, and Fig. 21e. During the $T_{inj}=50^\circ\text{C}$ scenario (Fig. 21c), the pressure behavior closely resembles that of the $T_{inj}=75^\circ\text{C}$ scenario, with minimal variation in the radial distance of the pressure front. However, during the injection of CO_2 at $T_{inj}=25^\circ\text{C}$, the pressure propagation results in the formation of two distinct pressure zones within the highly permeable layers, as depicted in Fig. 21e. In the region between these zones, conditions become favorable for CO_2 hydrate formation due to the combined effects of high pressure and significant cooling resulting from the JT cooling effect and conduction. The localized temperature drop in this area brings the fluid temperature close to or below the HFT, which triggers nucleation and subsequent growth of CO_2 hydrates. The formation of CO_2 hydrates in the pores can block the pore space and act as a permeability barrier, potentially impacting fluid flow and further influencing the pressure distribution within the reservoir.

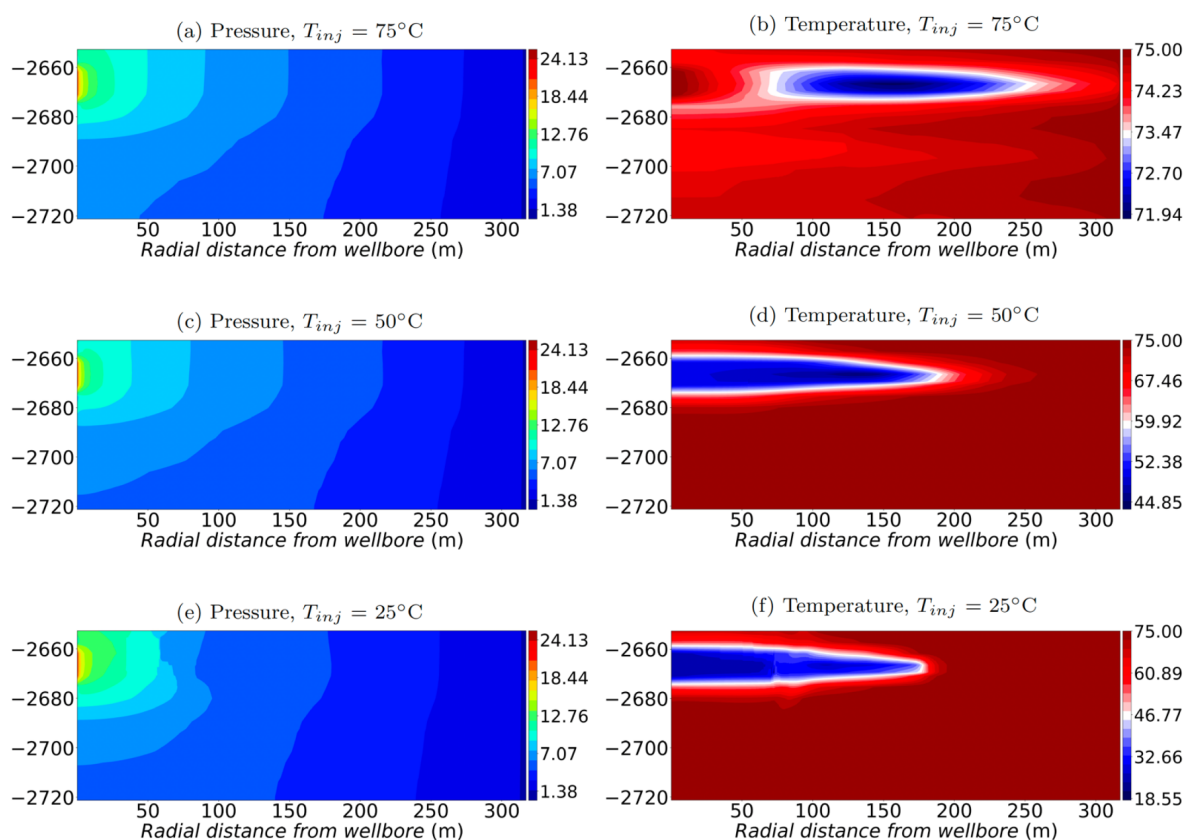


Figure 21—Dynamics of pore pressure and temperature for field test case-2 under different injection temperatures after 150 days.

The temperature variation is notable across all three scenarios of this field test case, as illustrated in Fig. 21b, Fig. 21d, and Fig. 21f. Near the injection well, the temperature matches the injection temperature (T_{inj}) of the injected CO_2 . The thermal front extends radially toward the outer boundary, accompanied by a gradual decrease in reservoir temperature. This effect is particularly significant in high-permeability layers due to elevated pressure gradients. Under these gradients, the Joule-Thomson (JT) cooling effect becomes prominent, accelerating CO_2 expansion and phase changes (see Fig. 1). In all scenarios, the low-temperature front propagates along layers 3 to 5, driven by the pressure gradient, as shown in Fig. 21b, Fig. 21d, and Fig. 21f. As previously discussed, the JT cooling effect is less significant when $T_{inj}=75^\circ\text{C}$, as depicted in Fig. 21b. The combined effects of heat transfer via conduction and JT cooling are more significant in the $T_{inj}=50^\circ\text{C}$ and $T_{inj}=25^\circ\text{C}$ scenarios. In the $T_{inj}=25^\circ\text{C}$ case, a notable pressure barrier and a temperature anomaly were observed. The propagation of low-temperature plumes in this scenario is primarily governed by heat transfer through conduction rather than the JT cooling effect. This behavior arises from the reduced pressure gradients following the formation of CO_2 hydrates in the porous media. The developed model effectively captures and analyzes these physical processes within the reservoir.

Conclusions

Most of the prior studies on Joule-Thomson cooling during the injection of CO_2 in depleted reservoirs have primarily concentrated on one-dimensional or steady-state models. This work aims to develop a coupled non-isothermal flow model that integrates hydrate formation kinetics and accounts for the dynamic properties of rock after the nucleation of CO_2 hydrates. The model is further integrated with the phase behavior of CO_2 hydrates and dynamic fluid properties to enhance its predictive capabilities concerning the dynamics associated with hydrate formation and fluid phase behavior.

The developed non-isothermal flow model is validated with analytical solutions to cross-check the accuracy of predicting dynamics of pore pressure, temperature, hydrate formation, fluid behavior, and its effect on the rock properties. The model is successfully implemented in the COMSOL multiphysics. The non-isothermal flow model results are in good agreement with the analytical model's steady state and transient solutions. The capabilities of the developed non-isothermal model were assessed using the Viking gas field. We conducted three test cases featuring different selections of layers, injection locations, and temperatures of 75°C, 50°C, and 25°C. These field cases served to implement the non-isothermal model, and the numerical model was built using radial coordinates. Suitable initial and boundary conditions were employed to observe the dynamics in pressure, temperature, hydrate formation, and fluid properties in these test cases.

The model effectively captures the dynamics of the JT cooling effect, which reduces temperature and triggers the nucleation of CO_2 hydrate, subsequently blocking the pores. This JT cooling effect is particularly significant in areas with high-pressure gradients, where the kinetics of hydrate formation are initiated in the model once the reservoir temperature reaches the hydrate formation temperature (HFT). Following hydrate formation, these pores become filled with CO_2 hydrate, reducing permeability and acting as a barrier, which impacts the pressure gradients observed in the pressure plots. This phenomenon diminishes the influence of the JT cooling effect and enhances heat transfer by conduction, as noted in the temperature plots. These complex dynamics are successfully captured by the developed non-isothermal flow model.

Understanding these processes is essential for optimizing CO_2 injection strategies, mitigating risks such as pore clogging and injectivity loss, and enabling the safe implementation of CO_2 sequestration projects under diverse operating conditions.

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Credit authorship contribution statement

Manojkumar Gudala: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization; **Amer Alanazi:** Writing – review & editing, Investigation, Formal analysis, Funding; **Marwan Fahs:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization; **Hussein Hoteit:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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